

DCG Description

Wizard Meng

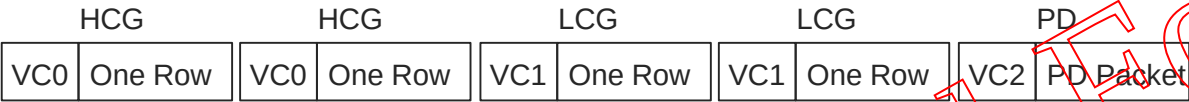
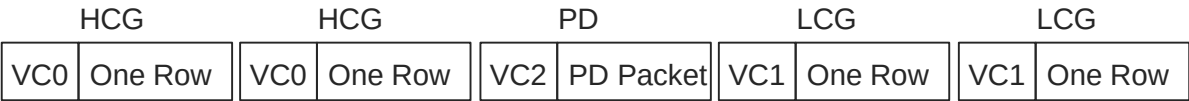
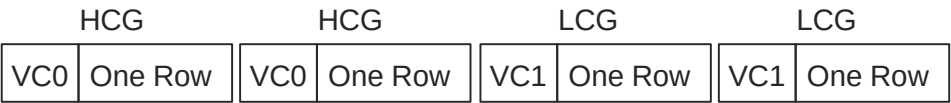
Safe Harbor Statement

- Except for historical information, the matters discussed in this presentation may contain forward-looking statements. These forward-looking statements are only estimates, and are subject to risks and uncertainties that may affect actual outcomes. Actual events or results may differ materially, and OMNIVISION disclaims any obligation to update information contained in any forward-looking statement.

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DCG output

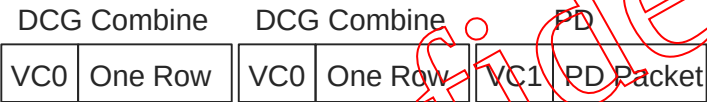
Without DCG combine



PD data may output during HCG H blanking or LCG H blanking.
For 4C2+ PD is from HCG. For 4C2 PD can be from HCG or LCG.

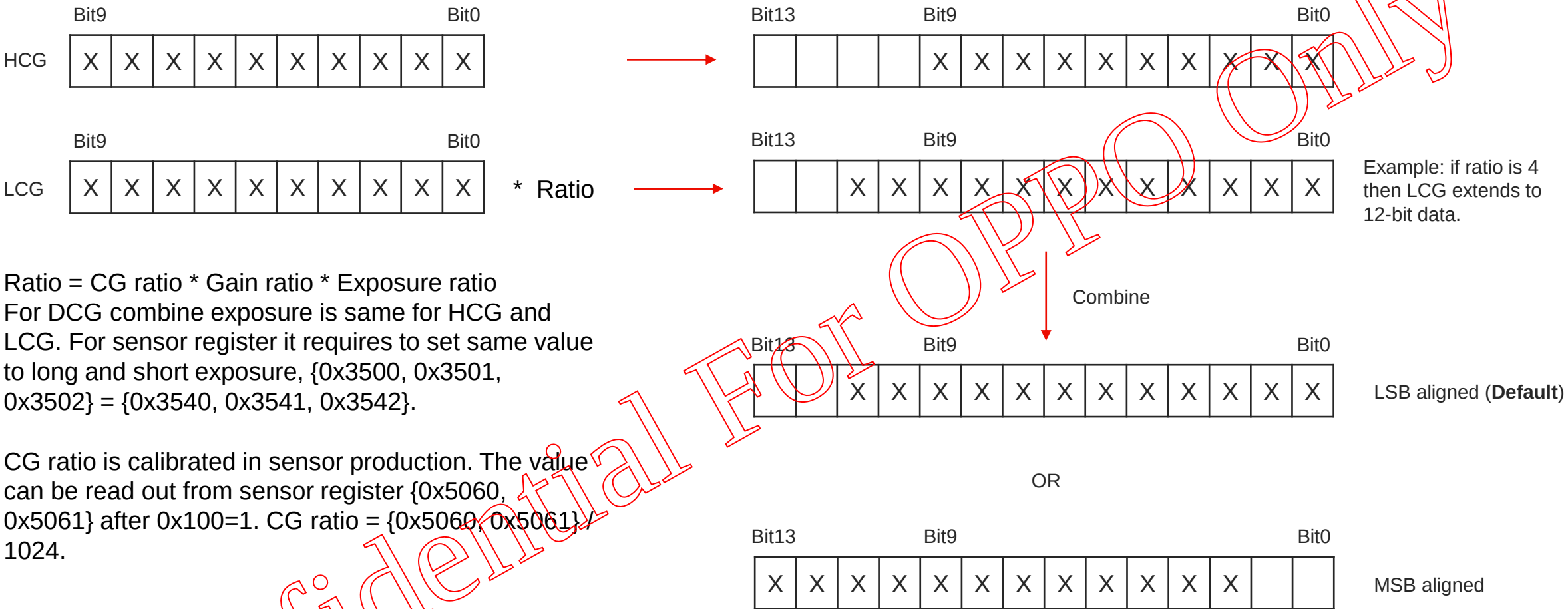
4C2+ has higher frame rate compare with 4C2. For some resolution and frame rate 4C2 has to limit LCG gain to achieve.
For example 4096x3072 30fps 4C2+ can do 64x on both HCG and LCG. 4096x3072 30fps 4C2 is HCG 16x and LCG 8x.

With DCG combine



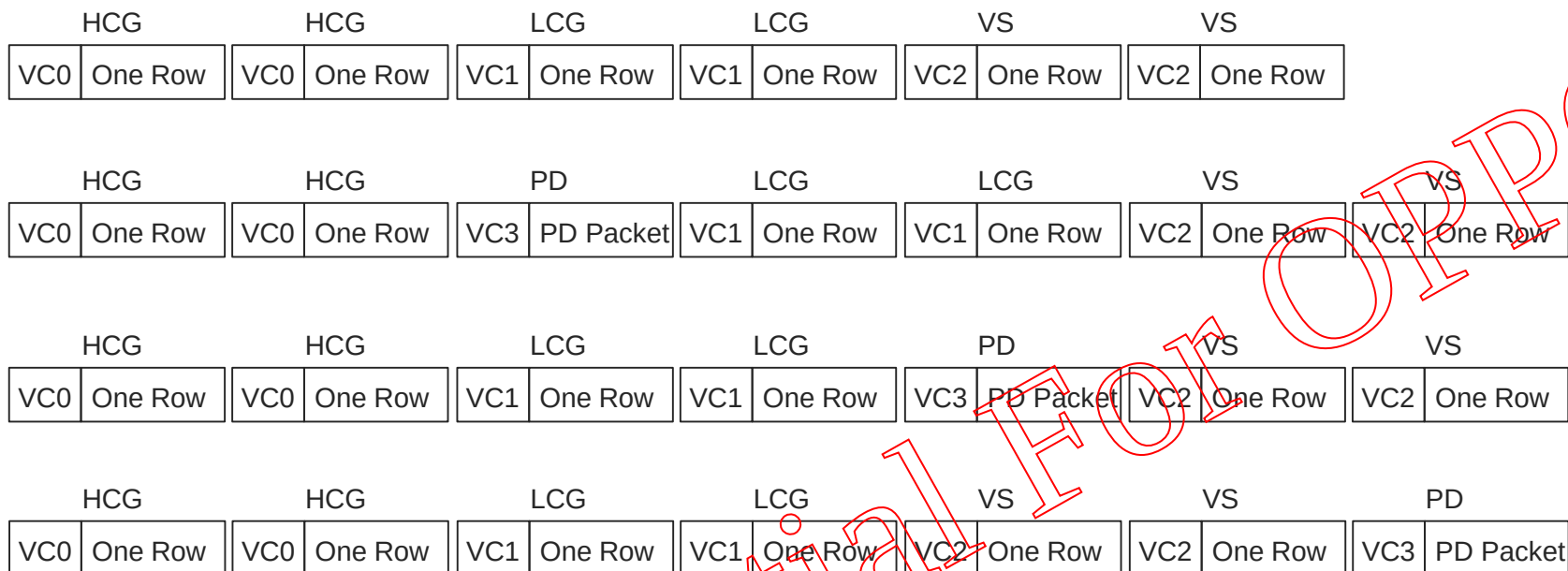
For 4C2+ PD is from HCG. For 4C2 PD can be from HCG or LCG.
It can be 10-bit or same as image bit width.

DCG Combine Data Alignment



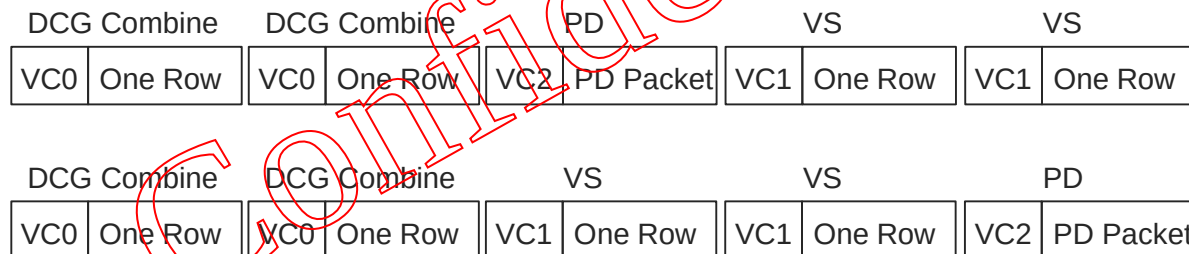
DCG + VS output

Without DCG combine



PD data may output during HCG H blanking or LCG H blanking or VS H blanking.
For 4C2+ PD is from HCG or VS.
For 4C2 PD is from HCG or LCG or VS.

With DCG combine



PD data may output during DCG combine H blanking or VS H blanking.
PD and VS can be 10-bit or same as image bit width.
For 4C2+ PD is from HCG or VS.
For 4C2 PD is from HCG or LCG or VS.

Data Bit Width

- Sensor combine is 14-bit but output can be 14-bit or 12-bit.

14-bit (default):

6c 3682 80

6c 468e 38

12-bit:

6c 3682 82

6c 468e 34

- PD/VS can be 10-bit or same as combined image bit width.

14-bit:

6c 0360 0b

6c 4681 88

12-bit:

6c 0360 0a

6c 4681 88

10-bit (default):

6c 0360 09

6c 4681 80

Thank You

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